n-Dimensional Regional Sampling

1 Setup Hoeffding and Bernstein Approach

We have a function $f : \mathbb{R}^p \to \mathbb{R}$ that is differentiable and has a bounded first derivative. We wish to estimate the integral $I = \int_{[0,1]^p} f(\boldsymbol{x}) d\boldsymbol{x}$.

1.1 Partition and Notation

1. Vectors: We write $\boldsymbol{x} = (x^{(1)}, \dots, x^{(p)})$.

2. Grid Points: For each dimension $k \in \{1, \dots, p\}$, we define partition points $\mathcal{P}_k = \{a_0^{(k)}, a_1^{(k)}, \dots, a_{m_k}^{(k)}\}$ where $0 = a_0^{(k)} < a_1^{(k)} < \dots < a_{m_k}^{(k)} = 1$.

3. The Index Set: Let $\mathcal{I} = \prod_{k=1}^{p} \{0, \dots, m_k - 1\}$ be the set of multi-indices $\mathbf{i} = (i_1, \dots, i_p)$.

4. Disjoint Regions: To ensure $[0,1]^p = \bigsqcup_{\mathbf{i} \in \mathcal{I}} \mathcal{A}_{\mathbf{i}}$, we define the intervals $J_{i_k}^{(k)}$ such that the right endpoint is included only at the boundary $x^{(k)} = 1$:

5. $$J_{i_k}^{(k)} = \begin{cases} [a_{i_k}^{(k)}, a_{i_k+1}^{(k)}) & \text{if } i_k < m_k - 1 \\ [a_{i_k}^{(k)}, 1] & \text{if } i_k = m_k - 1 \end{cases}$$

   The region is then $\mathcal{A}_{\mathbf{i}} = \prod_{k=1}^{p} J_{i_k}^{(k)}$.

1.2 Estimation

Denote the total sample budget as $N = \sum_{\mathbf{i} \in \mathcal{I}} N_{\mathbf{i}}$, where $N_{\mathbf{i}}$ is the budget for region $\mathcal{A}_{\mathbf{i}}$. We estimate the local and global integrals:

$$\int_{\mathcal{A}_{\mathbf{i}}} f(\boldsymbol{x}) d\boldsymbol{x} = \mathrm{Vol}(\mathcal{A}_{\mathbf{i}}) \cdot \mathbb{E}_{\boldsymbol{X} \sim \mathcal{A}_{\mathbf{i}}}[f(\boldsymbol{X})] \approx \frac{\mathrm{Vol}(\mathcal{A}_{\mathbf{i}})}{N_{\mathbf{i}}} \sum_{k=1}^{N_{\mathbf{i}}} f(\boldsymbol{X}_k), \quad \boldsymbol{X}_k \sim \mathcal{U}(\mathcal{A}_{\mathbf{i}})$$

Summing over all regions $\mathbf{i} \in \mathcal{I}$ yields the global estimator $\hat{I}$:

$$\hat{I} = \sum_{\mathbf{i} \in \mathcal{I}} \sum_{k=1}^{N_{\mathbf{i}}} \frac{\mathrm{Vol}(\mathcal{A}_{\mathbf{i}})}{N_{\mathbf{i}}} f(\boldsymbol{X}_k \sim \mathcal{A}_{\mathbf{i}})$$

1.3 Concentration Bound

We define $Z_k[\mathbf{i}] = \frac{\mathrm{Vol}(\mathcal{A}_{\mathbf{i}})}{N_{\mathbf{i}}} f(\boldsymbol{X}_k)$. The error probability is:

$$\mathbb{P}(|\hat{I} - I| > \varepsilon) = \mathbb{P}\left(\left|\sum_{\mathbf{i} \in \mathcal{I}} \sum_{k=1}^{N_{\mathbf{i}}} \{Z_k[\mathbf{i}] - \mathbb{E}[Z_k[\mathbf{i}]]\}\right| > \varepsilon\right) \tag{1}$$

Since $Z_k[\mathbf{i}]$ is bounded, it is sub-gaussian.

1.4 Hoeffding Approach

Applying the General Hoeffding inequality:

$$\mathbb{P}(|\hat{I} - I| > \varepsilon) \leq 2\exp\left(-\frac{c\varepsilon^2}{\sum_{\mathbf{i}\in\mathcal{I}}\sum_{k=1}^{N_{\mathbf{i}}}\|Z_k[\mathbf{i}] - \mathbb{E}[Z_k[\mathbf{i}]]\|_{\psi_2}^2}\right)$$

Using the centering lemma $\|Z_k[\mathbf{i}] - \mathbb{E}[Z_k[\mathbf{i}]]\|_{\psi_2}^2 \leq C\|Z_k[\mathbf{i}]\|_{\psi_2}^2$, we obtain:

$$\mathbb{P}(|\hat{I} - I| \geq \varepsilon) \leq 2\exp\left(-\frac{\tilde{c}\varepsilon^2}{\sum_{\mathbf{i}\in\mathcal{I}}\frac{\mathrm{Vol}(\mathcal{A}_{\mathbf{i}})^2}{N_{\mathbf{i}}}\|f\|_{\psi_2,\mathcal{A}_{\mathbf{i}}}^2}\right)$$

Hoeffding with max-min: Let $m_{\mathbf{i}}, M_{\mathbf{i}}$ be the min and max of f on $\mathcal{A}_{\mathbf{i}}$. The bound becomes:

$$\mathbb{P}(|\hat{I} - I| \geq \varepsilon) \leq 2\exp\left(-\frac{\tilde{c}\varepsilon^2}{\sum_{\mathbf{i}\in\mathcal{I}}\frac{\mathrm{Vol}(\mathcal{A}_{\mathbf{i}})^2}{N_{\mathbf{i}}}(M_{\mathbf{i}} - m_{\mathbf{i}})^2}\right)$$

1.4.1 Algorithm

The algorithm proceeds by iteratively refining the partition of $[0,1]^p$. Let d denote the current iteration (or depth).

Step 1: Initialization (Depth $d = 0$)
We begin with a single region covering the entire unit hypercube.

- **Partition Sets:** For each dimension $k \in \{1, \dots, p\}$, define the initial set of partition points $\mathcal{P}_k^{(0)} = \{a_0^{(k)}, a_{m_k}^{(k)}\} = \{0, 1\}$.
- **Dimensions:** $m_1 = m_2 = \cdots = m_p = 1$.
- **Index Set:** $\mathcal{I}^{(0)} = \{(0, 0, \dots, 0)\}$.
- **Initial Region:** $\mathcal{A}_{\mathbf{0}} = [0,1]^p$.

Step 2: Coordinate Splitting Selection
At each iteration, we evaluate the impact of splitting one coordinate. Suppose we consider splitting dimension j at a candidate point $t \in (a_{i_j}^{(j)}, a_{i_j+1}^{(j)})$. This split divides a target parent region $\mathcal{A}_{\text{parent}}$ into two new sub-regions:

- **Left Sub-region ($\mathcal{A}_j^-(t)$):** Inherits all interval definitions from $\mathcal{A}_{\text{parent}}$, except the j-th interval is updated to $[a_{i_j}^{(j)}, t)$.
- **Right Sub-region ($\mathcal{A}_j^+(t)$):** Inherits all interval definitions from $\mathcal{A}_{\text{parent}}$, except the j-th interval is updated to $[t, a_{i_j+1}^{(j)}]$.

Step 3: Optimization and Updating
We allocate samples n^+ to $\mathcal{A}_j^+(t)$ and $N - n^+$ to $\mathcal{A}_j^-(t)$. We calculate the local cost function $C_j(t)$:

$$C_j(t, n^+) = \frac{\mathrm{Vol}(\mathcal{A}_j^-(t))^2}{N - n^+}(M_j^-(t) - m_j^-(t))^2 + \frac{\mathrm{Vol}(\mathcal{A}_j^+(t))^2}{n^+}(M_j^+(t) - m_j^+(t))^2$$

where $M_j^{\pm}(t) = \max\{f(x) : x \in \mathcal{A}_j^{\pm}(t)\}$ and $m^{\pm}(t) = \min\{f(x) : x \in \mathcal{A}_j^{\pm}(t)\}$. Then it is easy to show that the optimal $n_*^+(t)$ as a function of t allocation as a ratio of N is given by

$$\frac{n_*^+(t)}{N} = \frac{\mathrm{Vol}(\mathcal{A}_j^+(t))(M_j^+(t) - m_j^+(t))}{\mathrm{Vol}(\mathcal{A}_j^-(t))(M_j^-(t) - m_j^-(t)) + \mathrm{Vol}(\mathcal{A}_j^+(t))(M_j^+(t) - m_j^+(t))}$$

Now we define the the objective function as $C_j(t) := C_j(t, n_*^+(t))$.

a. **Find Optimal Split:** For each j, find $t_j^* = \arg\min_t C_j(t)$.
b. **Select Coordinate:** Choose the dimension j^* that yields the minimum cost across all dimensions: $j^* = \arg\min_j C_j(t_j^*)$.
c. **Update Partition Set:** Update the set of points for the selected dimension:

$$\mathcal{P}_{j^*}^{(d+1)} = \mathcal{P}_{j^*}^{(d)} \cup \{t_{j^*}^*\}$$

d. **Re-index:** Increment m_{j^*} by 1. This update implicitly expands the multi-index set $\mathcal{I}$ and replaces the parent region with its two children in the global union.

Step 4: Iteration and Termination
The process repeats by applying Step 2 and Step 3 to the newly created sub-regions and we replace the number of samples $N \leftarrow n_*^+(t_{j^*}^*)$ for region $\mathcal{A}_j^+(t_{j^*}^*)$ and $N \leftarrow N - n_*^+(t_{j^*}^*)$ respectively. The algorithm terminates when:

- The maximum specified depth d is reached.
- The number of samples per region falls below a required threshold.

Remark 1. *The Hoeffding approach generally works better on function that behave sufficiently well so that $M_j^{\pm}(t)$ and $m_j^{\pm}(t)$ can be easily computed. We will talk about this more at the end.*

2 Bernstein Approach

From (1), we have that applying Bernstein inequality

$$\mathbb{P}(|\hat{I} - I| > \varepsilon) = \mathbb{P}\left(\left|\sum_{\mathbf{i}\in\mathcal{I}}\sum_{k=1}^{N_{\mathbf{i}}}(Z_k[\mathbf{i}] - \mathbb{E}[Z_k[\mathbf{i}]])\right| > \varepsilon\right) \leq 2\exp\left(-\frac{\varepsilon^2/2}{\sum_{\mathbf{i}\in\mathcal{I}}\sum_{k=1}^{N_{\mathbf{i}}}\mathrm{Var}(Z_k[\mathbf{i}]) + B\varepsilon/3}\right) \tag{2}$$

where $B > |Z_1[\mathbf{i}] - \mathbb{E}[Z_1[\mathbf{i}]]|$ almost surely for $\mathbf{i} \in \mathcal{I}$. One can take $B = M - m$ where $M = \max_{\boldsymbol{x}\in[0,1]^p} f(\boldsymbol{x})$ and $m = \min_{\boldsymbol{x}\in[0,1]^p} f(\boldsymbol{x})$. Thus, we can ignore $B\varepsilon/3$ and minimise the other term. We note that for any fixed $\mathbf{i} \in \mathcal{I}$, we have $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_{N_{\mathbf{i}}}$ are i.i.d, thus

$$\sum_{\mathbf{i}\in\mathcal{I}}\sum_{k=1}^{N_{\mathbf{i}}}\mathrm{Var}(Z_k[\mathbf{i}]) = \mathrm{Var}(\hat{I}) = \sum_{\mathbf{i}\in\mathcal{I}}\frac{\mathrm{Vol}(\mathcal{A}_{\mathbf{i}})^2}{N_{\mathbf{i}}}\mathrm{Var}(f(\boldsymbol{X} \sim \mathcal{U}[\mathcal{A}_{\mathbf{i}}]))$$

Thus the objective obtained from concentration recovers variance minimisation objective. There could be other works where an (greedy/adaptive/tree based) algorithm to determine the splitting boundary $t_{j^*}^*$ and $n_*^+(t_{j^*}^*)$ that minimises variance.

Equivalently the theoretical greedy optimisation step in the algorithm in Berstein approach turns to be

$$C_j(t) = \frac{\mathrm{Vol}(\mathcal{A}_j^-(t))^2}{N - n_*^+(t)}\mathbb{V}_j^-(t) + \frac{\mathrm{Vol}(\mathcal{A}_j^+(t))}{n_*^+(t)}\mathbb{V}_j^+(t), \qquad \frac{n_*^+(t)}{N} = \frac{\mathrm{Vol}(\mathcal{A}_j^+(t)) \cdot \sqrt{\mathbb{V}_j^+(t)}}{\mathrm{Vol}(\mathcal{A}_j^-(t)) \cdot \sqrt{\mathbb{V}_j^-(t)} + \mathrm{Vol}(\mathcal{A}_j^+(t)) \cdot \sqrt{\mathbb{V}_j^+(t)}}$$

where we use the shorthand $\mathbb{V}_j^{\pm}(t) = \mathrm{Var}_{\boldsymbol{X}\sim\mathcal{U}[A_j^{\pm}(t)]}(f(\boldsymbol{X}))$. One can note that we obtain the Neyman allocation for two strata as the optimal sample allocation $n_*^+(t)$. Now solving, for each dimension j, solving $t_j^* = \arg\min_t C_j(t)$ is computationally expensive as each step requires us to estimate $\mathbb{V}_j^{\pm}(t)$. Instead of treating t as a continuous variable, you restrict t to the co-ordinates of the samples currently inside that parent region. So we start by sampling N_{train} in $[0,1]^p$. Suppose a parent region currently contains n samples: $\{X_1, X_2, \ldots, X_n\}$. Sort these samples along dimension j so that $x_1^{(j)} \leq x_2^{(j)} \leq \cdots \leq x_n^{(j)}$. The only candidate split points t you evaluate are the midpoints between these existing samples: $t_k = \frac{x_k^{(j)} + x_{k+1}^{(j)}}{2}$ for $k = 1, \ldots, n-1$. We pick the midpoint t_j^* that minimizes the Bernstein cost. Then we follow the steps as in Hoeffding approach to find j^* and then we repeat until we have a tree of depth d (we specified at the start), and each time we inherit the samples of N_{train} from parent to child. There are some care needed here(but how?), when using N_{train} to learn how to split optimally and number of samples to allocate optimally we learn the optimal ratio of splitting as $n_*^+(t_{j^*}^*)/N_{\text{train}}$, when estimating we allocation $(n_*^+(t_{j^*}^*)/N_{\text{train}})N_{\text{est}}$ to $\mathcal{A}_{j^*}^+(t_{j^*}^*)$?